

Constants:

Fixed value. We can't change a constant value during program execution. Constant value should be provided at the time of declaration only. i.e. further initializations not allowed.

Numerical constants:

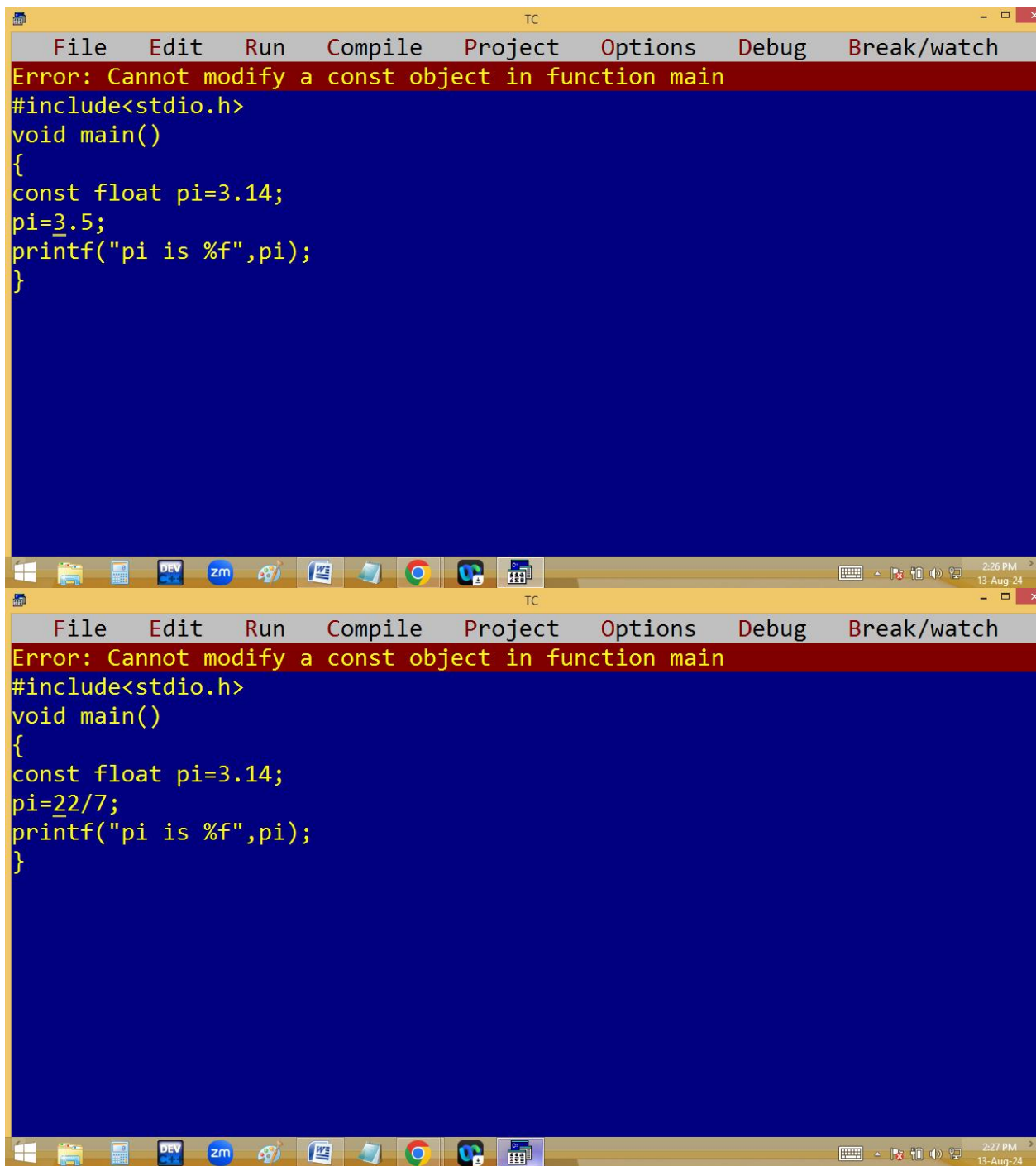
```
const float pi = 3.14;
```

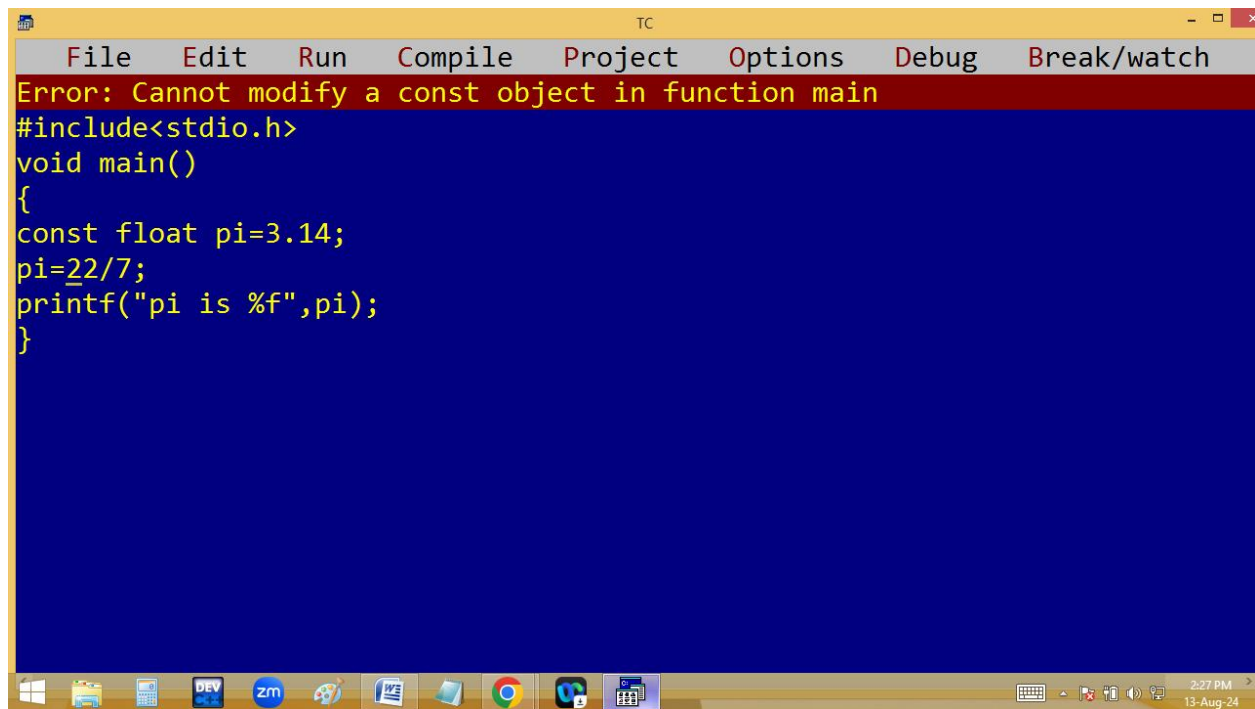
```
const int rollno=1234;
```

character constants:

```
const char name[ ]="Ravi"; string constant
```

```
const char gender='M'; ➔ char constant
```





The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". A red error message banner at the top reads "Error: Cannot modify a const object in function main". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
void main()
{
    const float pi=3.14;
    pi=22/7;
    printf("pi is %f",pi);
}
```

The Windows taskbar is visible at the bottom, showing icons for various applications including a file explorer, DEV C++, Zoom, and a web browser. The system clock in the bottom right corner indicates the time is 2:27 PM on 13-Aug-24.

The image shows two screenshots of the Turbo C++ (TC) IDE. The top window displays a compilation error for a C program. The code in the top window is as follows:

```
#include<stdio.h>
void main()
{
const float pi;
pi=3.14;
printf("pi is %f",pi);
}
```

The error message at the top of the window reads: "Error: Cannot modify a const object in function main".

The bottom window shows the same program after correction. The code is:

```
#include<stdio.h>
void main()
{
float pi; /* variable declaration */
pi=3.14; /* initialization */
printf("pi is %f",pi);
}
```

The status bar in the bottom window indicates "Line 5 Col 30 Insert Indent Tab Fill Unindent * E:2PM.C". The Windows taskbar at the bottom of both screenshots shows the time as 2:28 PM and 2:29 PM on 13-Aug-24.

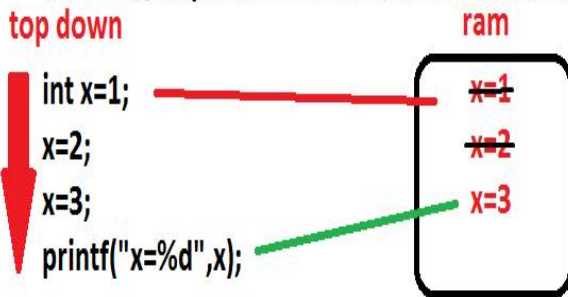
Identifiers:

Names of variables, functions, files, array, pointer etc.

Example for variables:

`int a=10;` to print 10 we have to call a then a is an identifier for 10

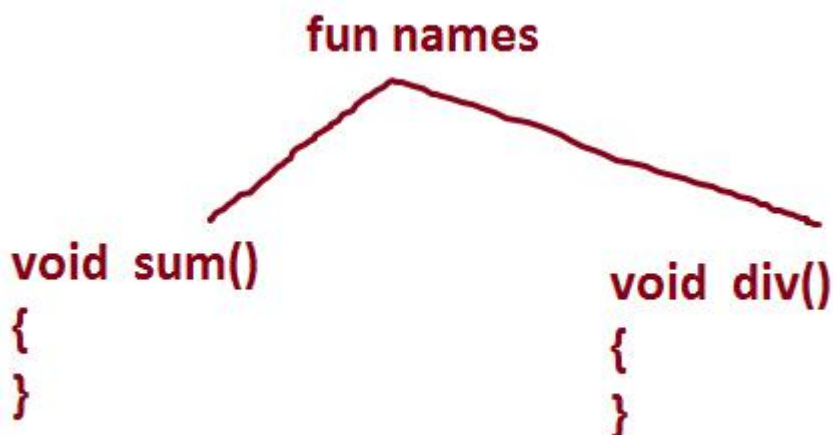
`int b=20;` to print 20 we have to call b then b is an identifier for 20



variable is a container is used to store the values in our programs

x is ready to change ==> vary means change and ready means able i.e. it is a variable

Example for functions:



Example for files:

a.c, b.cpp, c.java, d.py,.....

example for array:

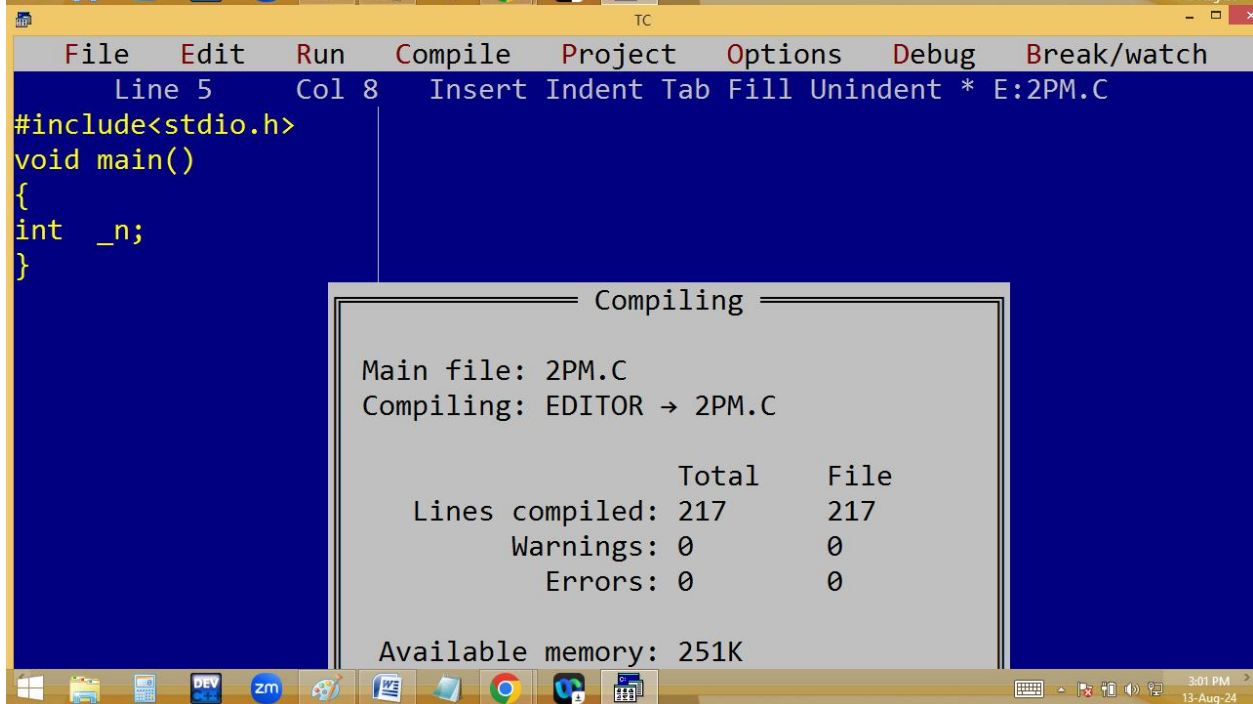
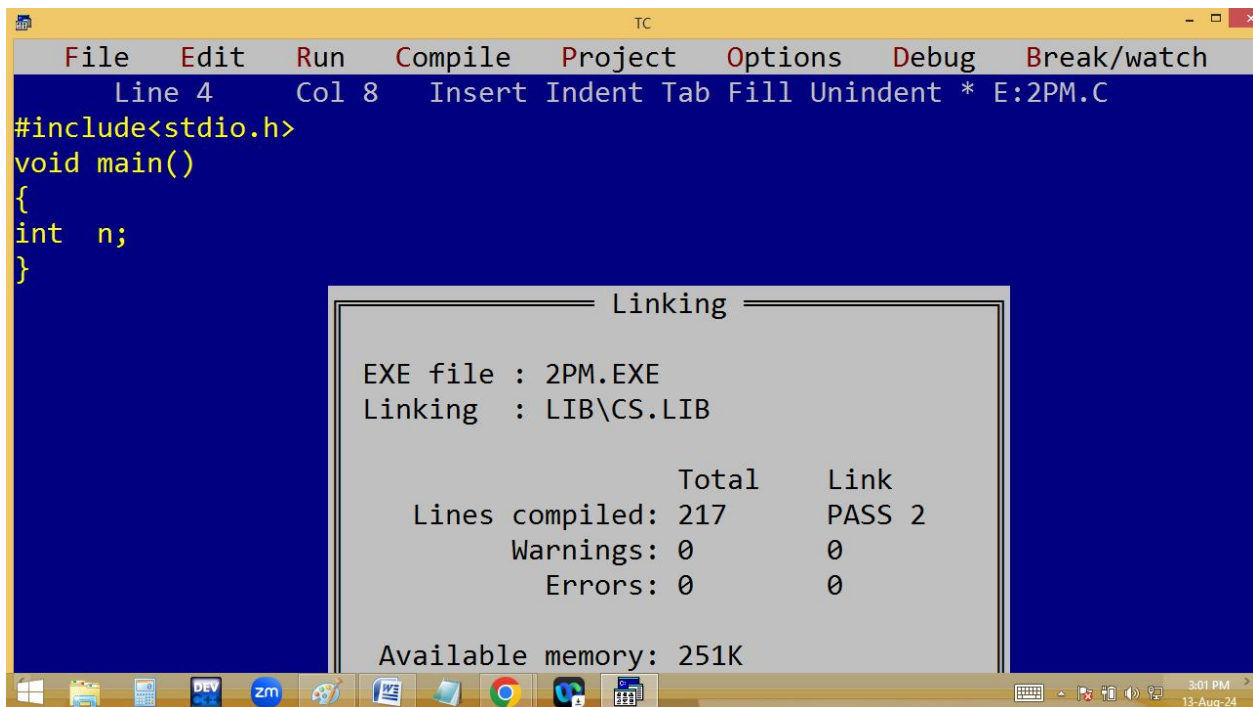
int a[10]; ← int array variable

example for pointer:

int * p; ← int pointer variable

Naming rules:

1. Name should have to start with alphabet or underscore[_].



The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 5, Col 7, Insert Indent Tab Fill Unindent * E:2PM.C). The code in the editor is:

```
#include<stdio.h>
void main()
{
int 1n;
}
```

A "Compiling" dialog box is open, displaying the following information:

Main file: 2PM.C
Compiling: EDITOR → 2PM.C

	Total	File
Lines compiled:	216	216
Warnings:	0	0
Errors:	1	1

Available memory: 250K

The Windows taskbar at the bottom shows the time as 3:01 PM on 13-Aug-24.

2. Numbers allowed but not at first position.

The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 4, Col 8, Insert Indent Tab Fill Unindent * E:2PM.C). The code in the editor is:

```
#include<stdio.h>
void main()
{
int n1;
}
```

A "Compiling" dialog box is open, displaying the following information:

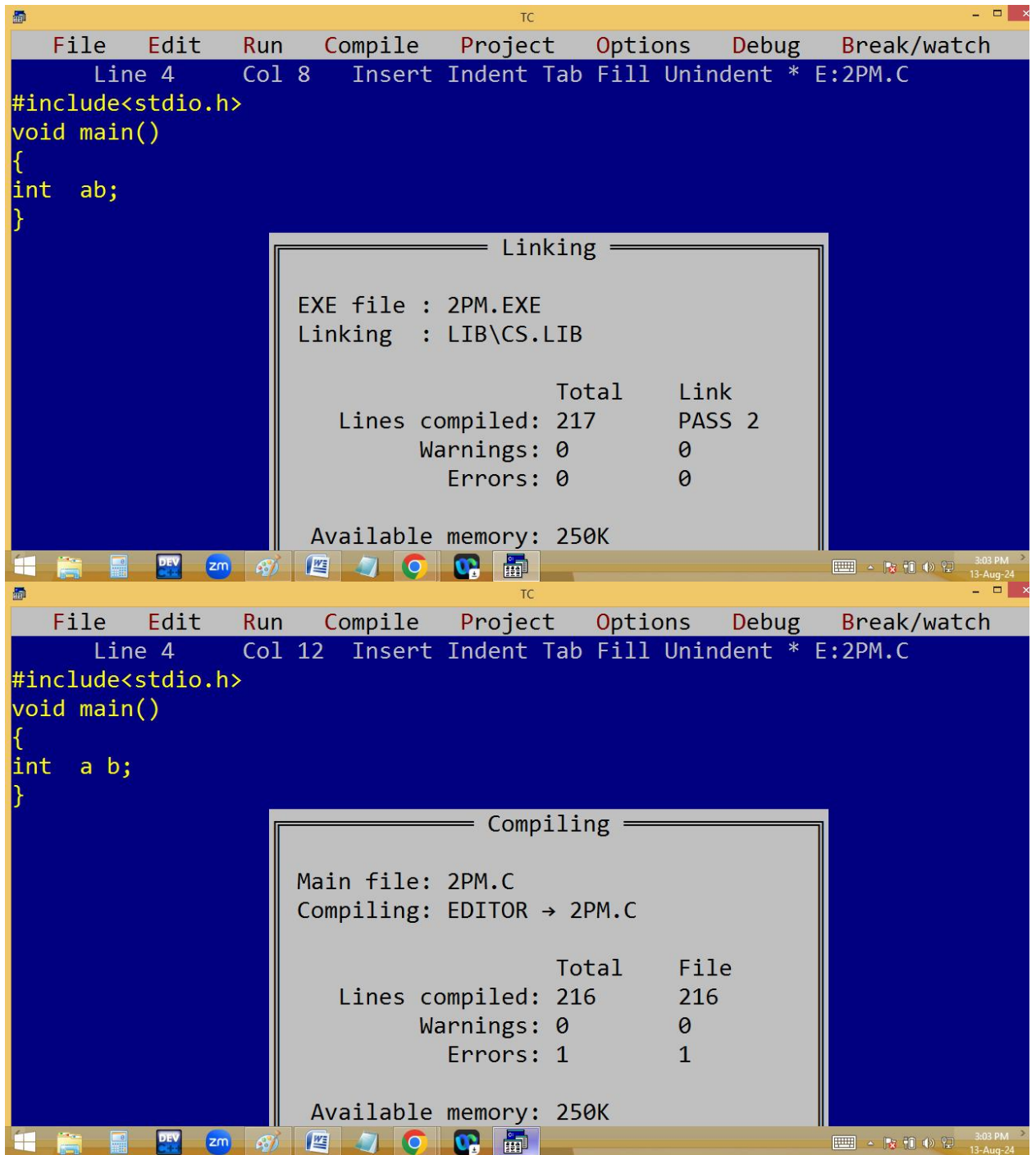
Main file: 2PM.C
Compiling: EDITOR → 2PM.C

	Total	File
Lines compiled:	217	217
Warnings:	0	0
Errors:	0	0

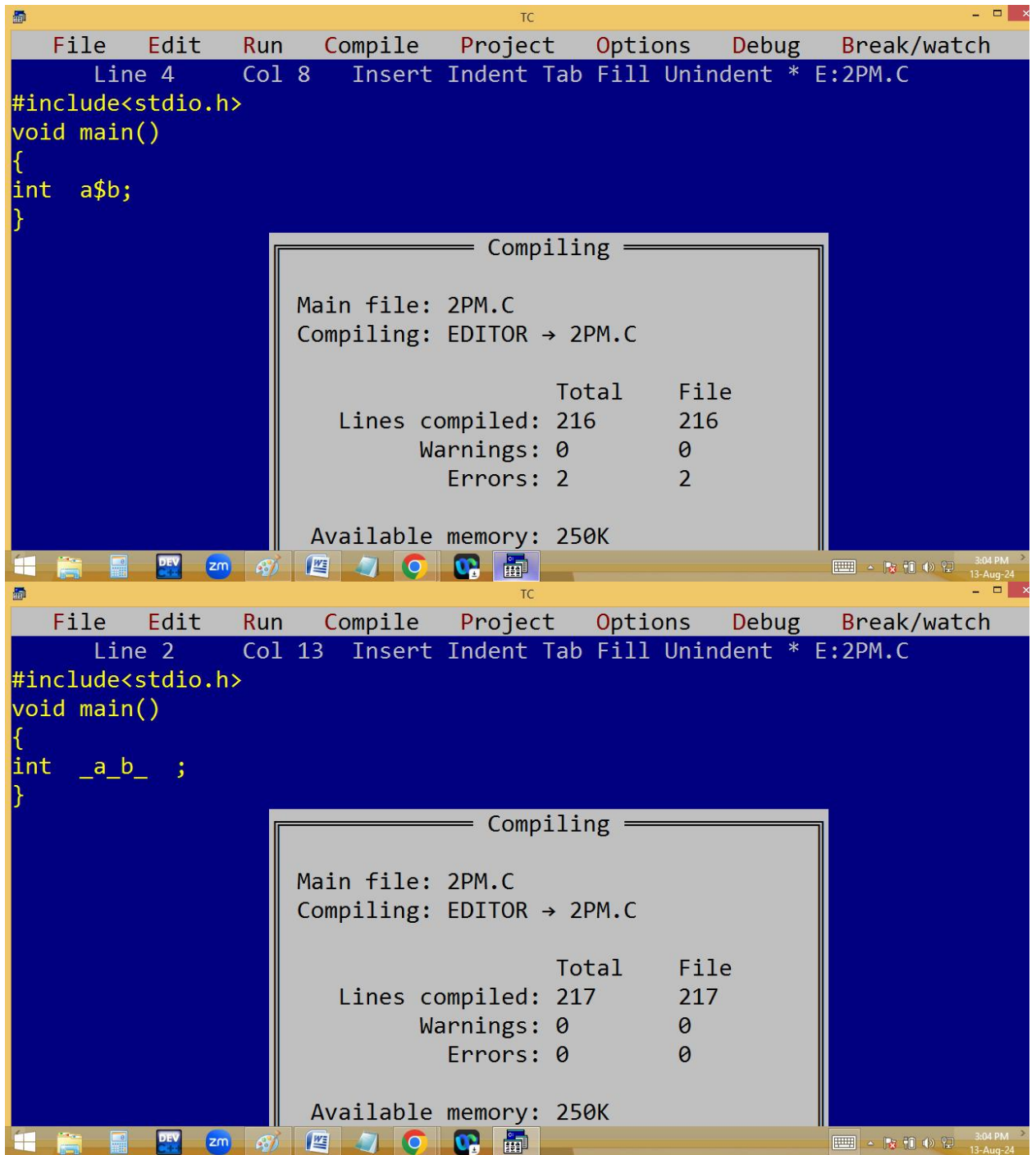
Available memory: 250K

The Windows taskbar at the bottom shows the time as 3:02 PM on 13-Aug-24.

3. Spaces not allowed.



4. No special characters except underscore.



5. Keywords not allowed as the identifiers

TC

File Edit Run Compile Project Options Debug Break/watch

Line 4 Col 11 Insert Indent Tab Fill Unindent * E:2PM.C

```
#include<stdio.h>
void main()
{
int float ;
}
```

Compiling

Main file: 2PM.C
Compiling: EDITOR → 2PM.C

	Total	File
Lines compiled:	216	216
Warnings:	0	0
Errors:	1	1

Available memory: 250K

Windows taskbar: 3:06 PM 13-Aug-24

TC

File Edit Run Compile Project Options Debug Break/watch

Line 4 Col 8 Insert Indent Tab Fill Unindent * E:2PM.C

```
#include<stdio.h>
void main()
{
int do ;
}
```

Compiling

Main file: 2PM.C
Compiling: EDITOR → 2PM.C

	Total	File
Lines compiled:	216	216
Warnings:	0	0
Errors:	1	1

Available memory: 250K

Windows taskbar: 3:06 PM 13-Aug-24

The screenshot shows the Turbo C++ (TC) IDE interface. The main window displays a C program with the following code:

```
Line 5 Col 7 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
void main()
{
int Do ;
}
```

A 'Linking' dialog box is open in the foreground, displaying the following information:

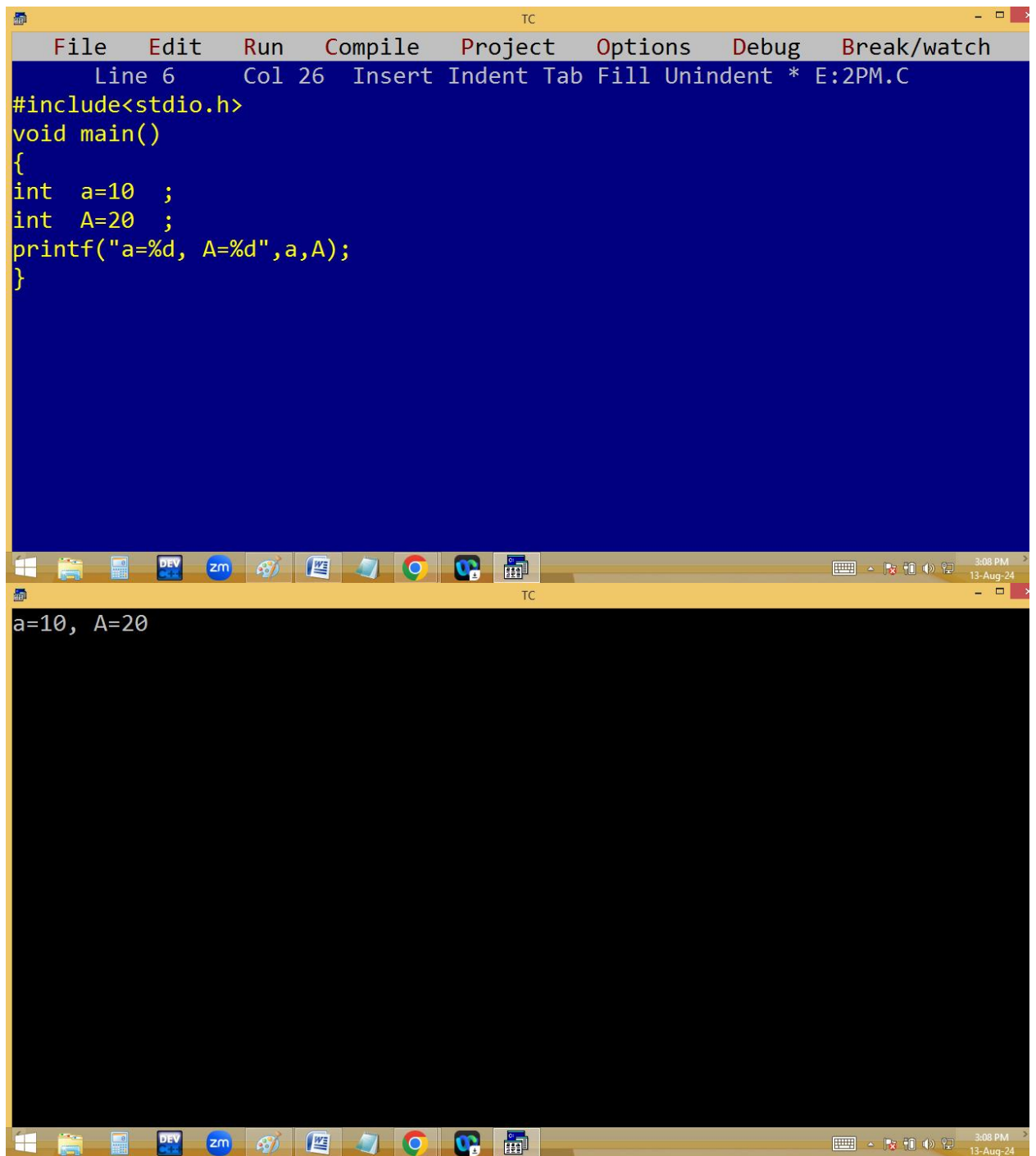
EXE file : 2PM.EXE
Linking : LIB\CS.LIB

	Total	Link
Lines compiled:	217	PASS 2
Warnings:	0	0
Errors:	0	0

Available memory: 250K

The Windows taskbar at the bottom shows the time as 3:06 PM on 13-Aug-24, along with various system icons and application icons.

6. Identifiers are case sensitive. i.e. lower and upper are different.



The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays a C program with the following code:

```
Line 6 Col 26 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
void main()
{
int a=10 ;
int A=20 ;
printf("a=%d, A=%d",a,A);
}
```

The bottom window shows the output of the program:

```
a=10, A=20
```

The IDE interface includes a menu bar with options: File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the bottom indicates the time as 3:08 PM on 13-Aug-24.

7. Duplicates names not allowed in same function or block { }. i.e. in different functions or block allowed.

TC

File Edit Run Compile Project Options Debug Break/watch

Error: Redeclaration of 'a' in function main

```
#include<stdio.h>
void main()
{
int  a=1  ;
int a=2  ;
int a=3  ;
printf("a=%d",a);
}
```

Windows Taskbar: File Explorer, DEV, zm, Paint, Word, Chrome, VS Code, Task View, 3:15 PM, 13-Aug-24

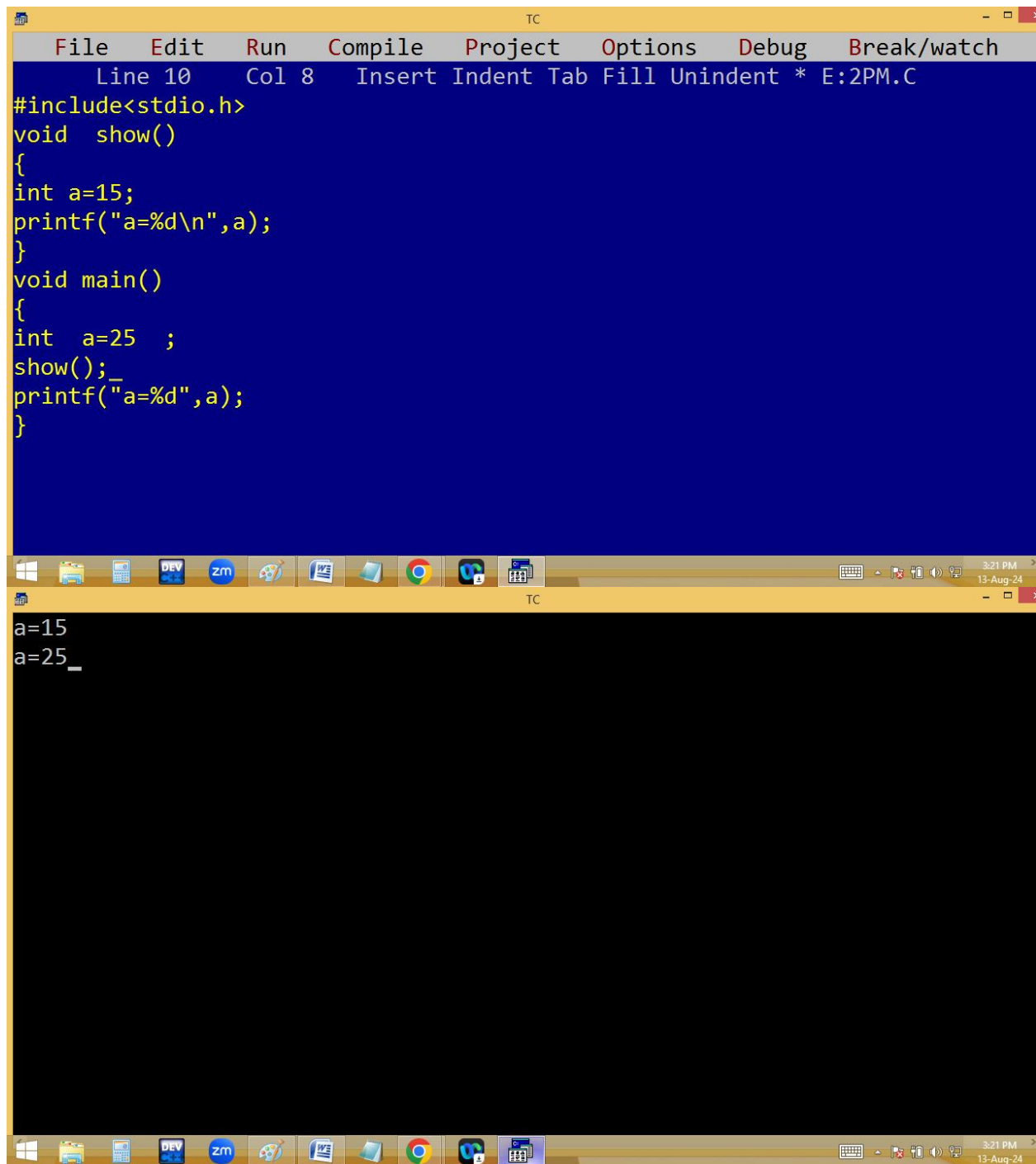
TC

File Edit Run Compile Project Options Debug Break/watch

Error: Redeclaration of 'a' in function main

```
#include<stdio.h>
void main()
{
int  a=10 ;
float a=2.5 ;
printf("a=%d",a);
}
```

Windows Taskbar: File Explorer, DEV, zm, Paint, Word, Chrome, VS Code, Task View, 3:16 PM, 13-Aug-24



The image shows two windows of the Turbo C++ (TC) IDE. The top window displays a C program with the following code:

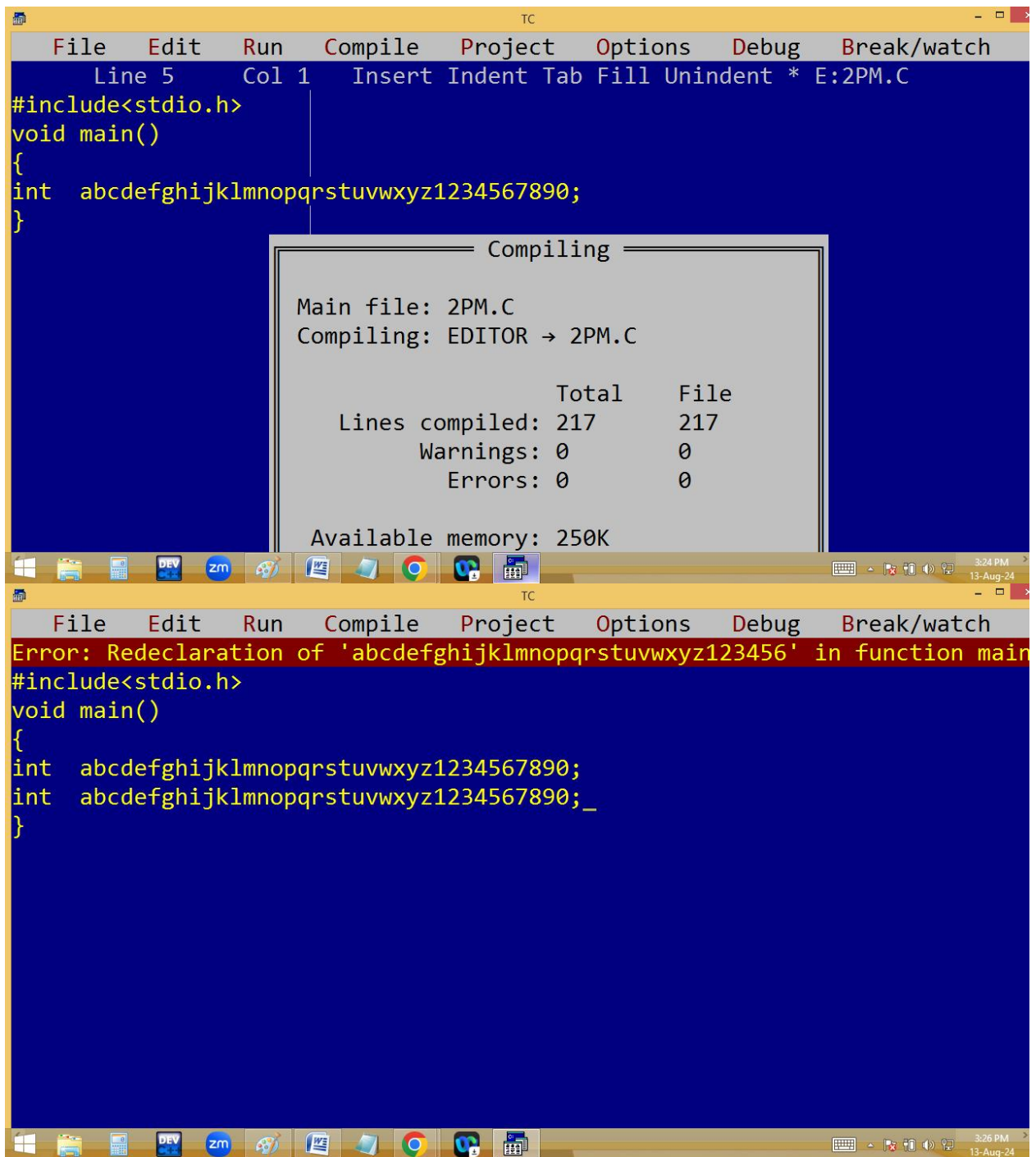
```
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 8 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
void show()
{
int a=15;
printf("a=%d\n",a);
}
void main()
{
int a=25 ;
show();
printf("a=%d",a);
}
```

The bottom window shows the output of the program:

```
a=15
a=25_
```

The IDE interface includes a menu bar at the top with options: File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. A status bar at the bottom right indicates the time as 3:21 PM on 13-Aug-24.

8. Name may contain up to 32 characters and excess characters ignored by the compiler.



The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot shows a successful compilation of a C program. The bottom screenshot shows the same program with a compilation error due to a redeclaration of a variable.

Top Screenshot: Successful Compilation

```
File Edit Run Compile Project Options Debug Break/watch
Line 5 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
void main()
{
int  abcdefghijklmnopqrstuvwxyz1234567890;
}
```

Compiling

Main file: 2PM.C
Compiling: EDITOR → 2PM.C

	Total	File
Lines compiled:	217	217
Warnings:	0	0
Errors:	0	0

Available memory: 250K

3:24 PM 13-Aug-24

Bottom Screenshot: Compilation Error

```
File Edit Run Compile Project Options Debug Break/watch
Error: Redeclaration of 'abcdefghijklmnopqrstuvwxyz123456' in function main
#include<stdio.h>
void main()
{
int  abcdefghijklmnopqrstuvwxyz1234567890;
int  abcdefghijklmnopqrstuvwxyz1234567890;_
}
```

3:26 PM 13-Aug-24

TC

File Edit Run Compile Project Options Debug Break/watch

Error: Redclaration of 'abcdefghijklmnopqrstuvwxyz123456' in function mai

```
#include<stdio.h>
void main()
{
int  abcdefghijklmnopqrstuvwxyz1234567890;
int  abcdefghijklmnopqrstuvwxyz123456;
}
```

TC

```
#include<stdio.h>
void main()
{
int  abcdefghijklmnopqrstuvwxyz1234567890;
int  abcdefghijklmnopqrstuvwxyz12345;
}
```

Linking

EXE file : 2PM.EXE
Linking : LIB\CS.LIB

	Total	Link
Lines compiled:	230	PASS 2
Warnings:	0	0
Errors:	0	0

Available memory: 250K

Success : Press any key

TC

Page: 10 of 17 Words: 177

3:27 PM 13-Aug-24

Page: 17 of 18 Words: 177

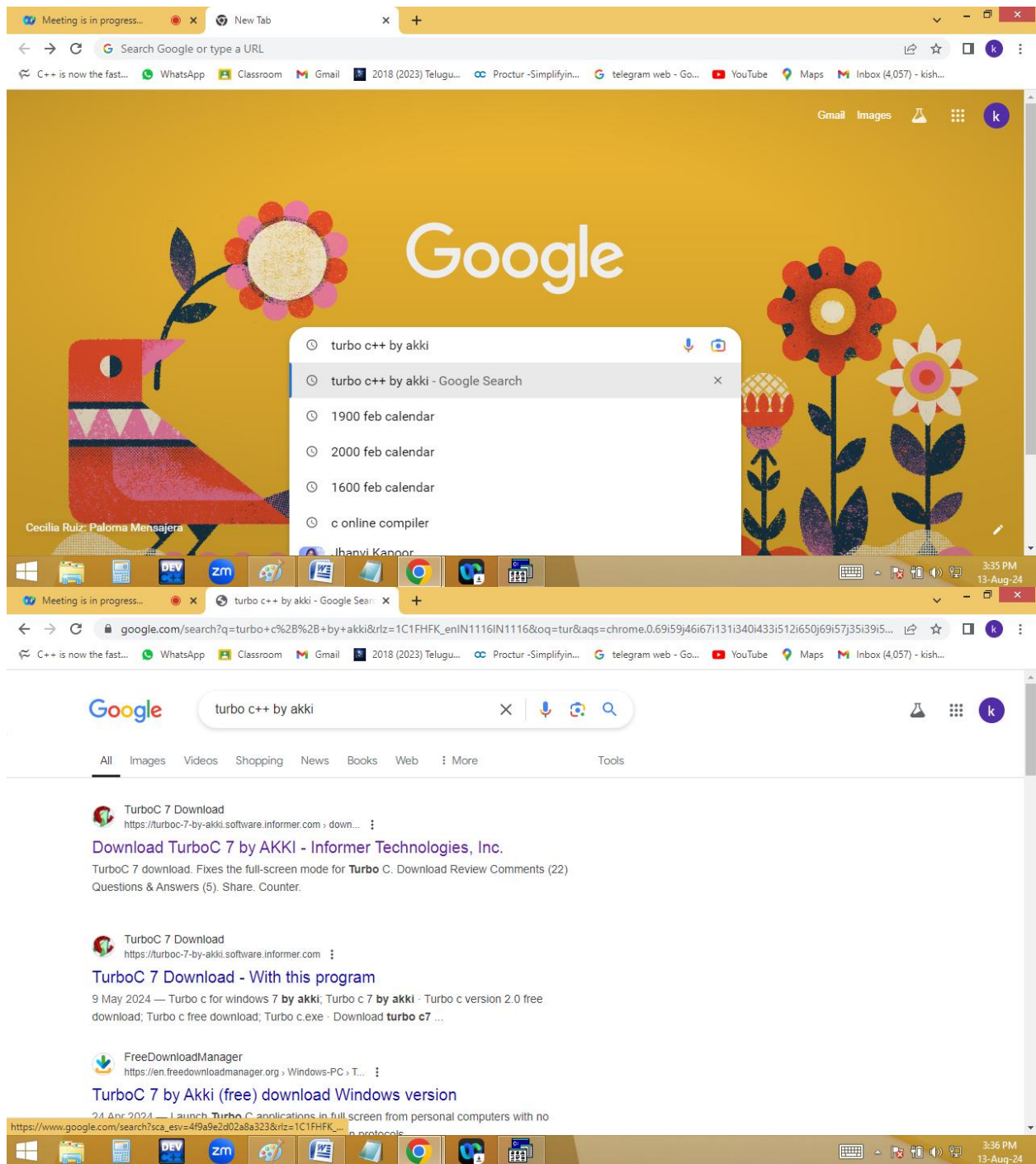
3:28 PM 13-Aug-24

Installing C software:

Open a browser [eg: chrome]

Type **turbo c++** by akki or turbo c++ for windows

[vscode/dev c++ / code blocks / free c / online compiler]



Click on download turboC 7 by akki

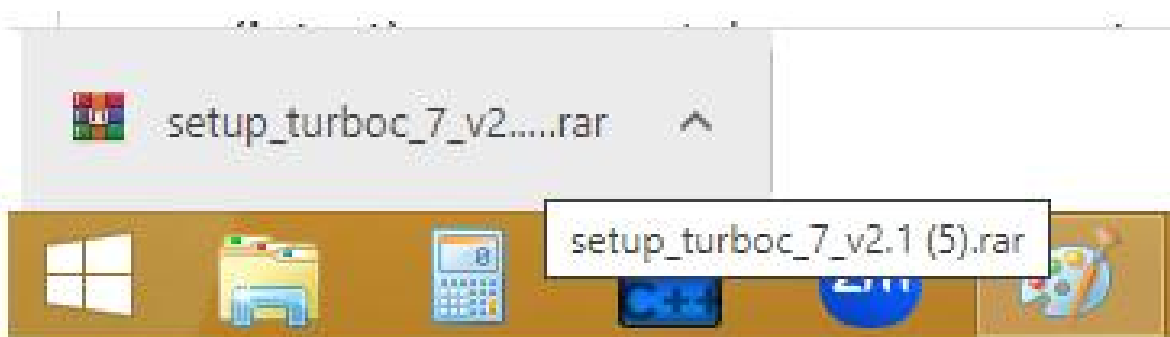
Or click on below link

<https://turboc-7-by-akki.software.informer.com/download/>

The screenshot shows a web browser window with the URL turboc-7-by-akki.software.informer.com/download/. The page has a navigation bar with tabs for 'Download', 'Review', 'Comments (22)', 'Questions & Answers (5)', and a 'SHARE' button. Below the navigation bar, there is an advertisement for 'Download Web Companion Free' on the left. The main content area features the title 'Download the latest version from Software Informer' with a green checkmark indicating the file is clean. A large blue 'Download now' button is prominent. To the right of the button, details are listed: 'Version: 2.1 (x86)', 'Date update: Jul 21, 2017', 'File name: setup_turboc_7_v2.1.rar', and 'Size: 3.4 MB'. Below the main content, there is a section for 'Latest versions of TurboC 7' showing a table with columns for version, date, and file name. The table lists '2.1 (latest)' as the current version, dated 'Jul 21, 2017', with the file name 'setup_turboc_7_v2.1.rar'. On the right side of the page, there is a 'Today's Highlight' section for 'CyberInstaller Suite'.

Version	Date	File Name
2.1 (latest)	Jul 21, 2017	setup_turboc_7_v2.1.rar

Click on download now button



Download any unzip software like winrar / 7zip etc.

Click on setup_turboc_7_v2....rar

Double click on **Setup_TurboC_7_v2.1.exe**

Click on yes → install → next → finish.